Discover: The Moon's Journey

Today I can… □ State that the Moon orbits Earth in roughly a month. □ Explain that moonlight is reflected sunlight. □ Explain phases using our changing view of the lit half.

Feedback: □ VF □ WS □ I □ NS □ E □ R □ [//] □ [?] _____

STAFF ANSWERS · Supported

Arrival

1. Full Moon: the visible disc is illuminated.
2. Planets orbit the Sun.
3. Sunlight reflected from the Moon.
4. Point to the visible part of the illuminated half.

Exit and independent answers

Check the pupil’s own evidence against the learning criteria. State that the Moon orbits Earth in roughly a month.; Explain that moonlight is reflected sunlight.; Explain phases using our changing view of the lit half.

Keep the first work and the changed work. A claim needs clues that fit and a science reason. Keep the model limit clear. These teaching examples do not show that you did a test.

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STAFF ANSWERS · Standard

Arrival

1. Useful: motion around a centre. Limit: scale or hand-driven motion.
2. Gravity.
3. Sunlight reaches the Moon and reflects towards the observer.
4. The Moon orbits Earth, changing our view of its illuminated half.

Exit and independent answers

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STAFF ANSWERS · Stretch

Arrival

1. An adult moving a model supplies the motion; it is not a measurement of gravity.
2. Orbit is travel around another body; rotation is turning on an axis.
3. Sunlight illuminates the Moon; reflected light travels towards the observer.
4. A phase changes our view of the lit half; a lunar eclipse blocks sunlight with Earth.

Exit and independent answers

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STAFF ANSWERS · teaching cues

Hinge

We see a changing part of the half lit by the Sun. This gives the normal phases. Earth's shadow matters in a lunar eclipse.

Model steps and limits

Torch is the Sun, ball is the Moon, observer is Earth. Keep the sunlight direction fixed.

Half the Moon is illuminated by the Sun. As it orbits, our view of the illuminated half changes.

When the visible lit half faces us, we see full Moon. When it faces away, we see new Moon.

Trace the light path: Sun → Moon → observer on Earth. Name the Sun as light source and the Moon as reflecting object.

The Moon orbits Earth in about 27.3 days, roughly a month. The phase cycle is about 29.5 days.

Sketch the view beside each physical model position. Mark the sunlight direction.

A lunar eclipse occurs when Earth blocks sunlight from reaching the Moon; the tilted orbit means this is not monthly.

The model does not show realistic sizes, distances or the tilt of the Moon’s orbit.

Activity answers

Crescent: A crescent is our view of part of the sunlit half.

Full Moon: Full Moon is an ordinary phase; it does not require Earth’s shadow.

New Moon: At new Moon the visible side is largely unlit.

Dark red Moon: A lunar eclipse involves Earth’s shadow.

Own light: Moonlight is reflected sunlight.

Every month: The Moon’s tilted orbit prevents an eclipse every month.

Misconception repair

In science, a claim needs clues that fit. Give a reason. Say how far the claim can go.

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STAFF ANSWERS · teaching cues · continued

Moon statement answers

The Moon reflects sunlight. True: the light reaching us is reflected sunlight.

The Moon glows by itself. False: it reflects sunlight rather than making its own visible light.

Ordinary phases are Earth’s shadow. False: phases come from our changing view of the lit half; Earth’s shadow matters during a lunar eclipse.

The Moon orbits Earth in about a month. True: about 27.3 days for one orbit; the phase cycle is about 29.5 days.

The Moon is only visible at night. False: the Moon can also be visible in daylight.